

Introduction to Programming II Project Log

Project title:	Music Visualizer
Topic:	Skull Sonic Vibes
Week	6-7
What progress have you made on this topic?	
Developed a JavaScript visualizer named "Skull Sonic" that dynamically renders a grid of skulls on a canvas. Created a Skull function serving as a constructor for initializing properties (name, x, y, amplitude) and methods (draw, drawOne, drawTwo, drawThree). Implemented a graphical grid system using nested loops to dynamically render multiple skull designs (drawOne, drawTwo, drawThree) across the canvas. Integrated dynamic scaling of skull sizes based on audio amplitude levels, ensuring visual responsiveness to music inputs. Utilized switch-case statements to handle user keyboard inputs (A, B, C) for real-time style switching of the visualizer.	
What problems have you faced and were you able to solve them?	
Challenge: Ensuring smooth integration of amplitude scaling without compromising performance. Solution: Used <code>getLevel()</code> method to accurately map amplitude values to skull sizes, optimizing performance with <code>scale()</code> function.	
What are you planning to do over the next few weeks?	
Refine skull designs for aesthetic variety and visual appeal. Implement additional interactive features such as color variations or animation effects based on user input.	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
Currently on track to complete core functionality and initial feature set within the planned timeline. If deviations arise, will prioritize bug fixes and optimization to maintain project momentum and quality standards.	

Project title:	Music Visualizer
Topic:	Musical Ballerina
Week	7-8
What progress have you made on this topic?	
Developed Ballerina class to animate musical ballerina. Implemented initial properties: name, x, y, numPoints, radius. Created draw method to render music representation and path. Implemented dance method to animate ballerina movement based on circular path. Utilized WebGL for efficient graphics rendering and enhanced animation capabilities. Integrated FFT (Fast Fourier Transform) for audio spectrum analysis and music visualization. Added multiple ballerina profiles (drawOne, drawTwo, etc.) for visual variety.	
What problems have you faced and were you able to solve them?	
Issue: Difficulty making the ballerina move on a circular path. Solution: Implemented trigonometric calculations in the dance method to ensure smooth circular motion. Issue: Challenge in implementing dynamic profile changes for the ballerina. Solution: Developed a profile switching mechanism	
What are you planning to do over the next few weeks?	
Refine ballerina profiles for visual appeal and thematic coherence. Implement interactive features for user engagement (e.g., change ballerina style, music selection). Conduct extensive testing for performance, visual appeal, and user experience feedback.	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
On Target: Yes. If Off Target: Adjust animation or profile switching logic for smoother transitions.	

Project title:	Music Visualizer
Topic:	Coffee Cup
Week	9
What progress have you made on this topic?	
Significant progress has been made on implementing the "Coffee Cup" visualizer using p5.js. The core functionality includes a dynamic coffee cup that bounces in response to audio amplitude, synchronized with changing background images. The CoffeeCup function encapsulates methods for drawing the cup, managing background transitions, and animating smoke trails using sine wave patterns. Each visual element scales and changes dynamically based on real-time audio input, enhancing the immersive experience. Integration of these components has resulted in a cohesive and interactive musical visualization framework.	
What problems have you faced and were you able to solve them?	
Initially, aligning the smoke animation with audio fluctuations posed a challenge. Adjustments in the animation loop and fine-tuning of amplitude mappings resolved synchronization issues, ensuring smooth and responsive visual feedback to varying music dynamics.	
What are you planning to do over the next few weeks?	
In the coming weeks, I aim to enhance the visualizer's interactivity and aesthetics. This includes refining animation transitions, optimizing code, and selecting some better fitting background images.	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
The project is progressing on schedule towards completion. To ensure timely delivery, I will focus on completing core functionalities first, prioritizing refinement and optimization thereafter. Regular milestones will be set to monitor progress and address any unforeseen challenges promptly.	

Project title:	Music Visualizer
Topic:	Harmony of Branches
Week	10
What progress have you made on this topic?	
Significant progress has been made in developing the FractalTree constructor function for the music visualizer. The code now efficiently renders fractal tree structures on a canvas, dynamically adjusting branch sizes and angles based on audio input. The draw() method initializes the canvas, assigns random stroke colors for visual variety, and recursively draws branches using amplitude values obtained from audio analysis. The branches(len) function has been optimized to handle recursive calls effectively, ensuring smooth and responsive animation	
What problems have you faced and were you able to solve them?	
Addressing performance concerns with recursive rendering was a primary challenge. By fine-tuning recursion parameters and implementing caching mechanisms for rendered branches, I achieved notable improvements in animation fluidity and responsiveness. Integrating real-time audio analysis to dynamically adjust tree parameters required refining the getLevel().	
What are you planning to do over the next few weeks?	
Conducting extensive testing across diverse music genres and audio sources to validate robust performance and visual appeal under varying conditions. Documenting code comprehensively. After Midterms: I plan to further develop this extension after the midterms, enhancing it so that the branches will expand and retract dynamically with the music.	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
On Target: Yes. If Off Target: If deviations arise, will prioritize bug fixes and optimization to maintain project momentum and quality standards.	

Project title:	Music Visualizer
Topic:	Dancing Emotions
Week	13 - 14
What progress have you made on this topic?	
I have made significant progress in developing the "Dancing Emotions" visualizer with p5.js. The main feature is a central disco ball that randomly changes colors and mirrors the dancing movements of five stickmen, each symbolizing a distinct emotion. These stickmen react to specific frequency ranges, causing jumps and spotlight effects when certain audio frequencies are recognized. The combination of real-time frequency analysis with visual components like jump states and highlight effects has resulted in a dynamic and engaging visual experience.	
What problems have you faced and were you able to solve them?	
A major problem I faced was precisely mapping the frequency ranges to the relevant stickmen motions while also ensuring that the lighting effects synchronized with their movements. Initially, the frequency detection was not accurate enough, resulting in inaccurate jump triggers. I resolved this by adjusting the frequency range values.	
What are you planning to do over the next few weeks?	
In the coming weeks, I intend to begin work on my prior extension, "Harmony of Branches." This is a fractal tree that currently scales and maps branch angles according to music. However, I am going to start updating it to align with the specifications and improvements outlined in my report..	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
On Target: Yes.	

Project title:	Music Visualizer
Topic:	Harmony of Branches - final version
Week	15 - 16
What progress have you made on this topic?	
I've made notable modifications to the "Harmony of Branches" extension since my midterms. The first version of the extension, submitted for the midterms, featured a visualization of a fractal tree with branch size and angle controlled by amplitude. The revised version now includes more features. I've added a maxDepth variable to vary the depth of the branches based on frequency energy. Additionally, I've improved the branch function to handle variations in branch depth and length based on high-frequency energy and amplitude. The fractal tree branches now extend and contract in accordance with the music amplitude value through updates to the branches and draw functions.	
What problems have you faced and were you able to solve them?	
The primary challenge I faced was adding the maxDepth feature and ensuring it worked properly with the branches function. Initially, mapping the high-frequency energy to regulate the branches' depth caused the visualization to lag. I was able to fix this by making some adjustments in the map function.	
What are you planning to do over the next few weeks?	
In the following weeks, I plan to focus on testing the present implementation to ensure that all functionalities work properly and effectively. I will begin working on the project's next extension, which might involve adding more visually appealing extensions to the primary template and using color theory to improve the present visual effects	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
On Target: Yes.	

Project title:	Music Visualizer
Topic:	Hanging Lines
Week	18 - 19
What progress have you made on this topic?	
I successfully constructed the "Hanging Lines" visualizer using p5.js for real-time audio visualization. The primary feature consists of displaying 10 lines with dynamically changing lengths and gradients based on different parts of the audio frequency spectrum. Circular components are also created at the bottom of each line, with their sizes matched to the music's amplitude. The approach efficiently mixes the movement of lines and circles with gradient effects to produce a visually appealing and synchronized audio presentation.	
What problems have you faced and were you able to solve them?	
One of the most difficult challenges I faced was precisely mapping the frequency spectrum values to line lengths, ensuring that the visual response was smooth and in tune with the music. The early implementation caused sudden changes in line length, disrupting the visual flow. I resolved this by fine-tuning the mapping function and adjusting the frequency gap.	
What are you planning to do over the next few weeks?	
In the next few weeks, I plan to thoroughly test the existing visualizer to identify and fix any errors or performance problems. I will also start designing and implementing the next project extension along with testing.	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
On Target: Yes.	

Project title:	Music Visualizer
Topic:	Vintage Retro Cassette
Week	16 - 17
What progress have you made on this topic?	
The nostalgic aesthetics of a vintage cassette player inspired me to create the "Vintage Retro Cassette" visualizer. It features a detailed cassette body design, rotating decks that react to the bass levels in the music, an animated musical note that moves in sync with the music amplitude, and a frequency graph that visualizes the audio waveform with a smoothing effect. The tight integration of these visual components with the audio input creates an interesting visual experience.	
What problems have you faced and were you able to solve them?	
One of the most difficult problems I faced was precisely mapping the frequency spectrum values to line lengths, ensuring that the visual response was smooth and in tune with the music. The early implementation caused sudden changes in line length, disrupting the visual flow. I handled this by fine-tuning the mapping function and adjusting the frequency gap.	
What are you planning to do over the next few weeks?	
Over the next two weeks, I'm going to concentrate on testing to make sure everything is operating as it should. I'll also adhere to my Gantt chart, which shows the precise times I've allotted for each assignment. Additionally, I'll start compiling the documents needed for the project's documentation.	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
On Target: Yes.	

Project title:	Music Visualizer
Topic:	Rectangle
Week	19 - 20
What progress have you made on this topic?	
I've made significant progress on the "Rectangle Animation" extension this week. The main features of the Rectangle function, including initializing rectangle properties like baseSize, sizeIncrement, minDelay, and maxDelay, have been successfully implemented. Rectdraw, which controls the formation and display of rectangles based on the amplitude of the music, is called by the draw function. The timer variable ensures that rectangles are added at the proper intervals, while the map functions dynamically map the size and opacity of the rectangles to the amplitude.	
What problems have you faced and were you able to solve them?	
I haven't faced many issues with this extension; the only problem was adding the timer to control the frequency of rectangle rendering. I resolved this by adjusting the timerDelay to better synchronize with the music amplitude. Additionally, mapping the size and opacity of the rectangles was straightforward.	
What are you planning to do over the next few weeks?	
Over the next few weeks, I plan to gather all of my project logs to finish the project documentation. I'll complete the testing phase and start working on the report as soon as the documentation is complete.	
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?	
On Target: Yes.	